Panel Q&A: likely questions and honest answers

Short answers to say out loud, each with where the evidence is. "Q" and "PI" numbers are the clinic interview (`docs/defense/INTERVIEW_EVIDENCE.md`). Test files are under `backend/tests/Feature/` unless they start with `tests/`.

A. Business process automation

1. What business process does the system automate? The patient's path from booking to receipt: online booking without conflicts, confirmation emails, front-desk messaging, invoices generated from the treatment, online payment settled automatically from PayMongo, and the Payment Receipt emailed with a PDF. Evidence: `docs/defense/AUTOMATION_EVIDENCE.md`.

2. Which manual steps disappear for the receptionist? Typing bookings and checking conflicts (Q20 says scheduling conflicts are the most frequent error), recording online payments and preparing receipts (Q27: booking, billing and follow-up are done by the receptionist), and answering routine questions (PI-43: the Owner answers Messenger herself).

3. How do you prevent double booking? Every booking passes the same server checks (branch open that day, Dentist eligible and free, patient not double-booked), with a database-level guard against two people taking the same time at once. `Appointments/ConcurrentBookingTest`, `AppointmentBookingTest`.

4. Does it shorten the queue and waiting time (Q19, Q21, Q32)? Indirectly: patients arrive with a fixed booked time, and the queue module numbers arrivals. We have not measured waiting times at the clinic, so we do not claim a percentage.

5. What happens when the patient pays online? PayMongo sends a signed webhook; the server verifies it, marks the invoice paid, creates the receipt and emails the PDF. The return page alone never marks anything paid. `Payment/PayMongoPaymentTest`.

6. Is the chatbot AI? It is an automated assistant that answers from the clinic's own approved information and records (knowledge mode). Nothing a patient types leaves the clinic's system. An optional AI mode exists for a clinic that supplies its own Anthropic key and approves the data processing. `Assistant/AssistantKnowledgeTest`, `docs/ASSISTANT_PRIVACY.md`.

7. What if the assistant doesn't know the answer? It says so and offers the front desk; it never makes up facts, prices or policies. Tested with unknown questions (parking, walk-ins).

8. Can the assistant book or pay for the patient? No. It only shows buttons (open booking, open billing, message the front desk). It refuses medical advice and sends emergencies to the clinic phone.

B. Security and privacy (RA 10173)

9. How are patients' records protected? Role-based access (Patient, Staff, Dentist, Owner) checked on the server for every request; Staff see only their branches; a patient sees only their own data (tested: another patient's invoice answers "not found"). `RbacTest`, `Billing/BillingDocumentDownloadTest`, `Reference/ReferenceDataAuthorizationTest`.

10. How do you verify identity? Email verification with a 6-digit code (10 minutes, limited attempts), password reset by code, current password required to change the email, and a security notice to the old address. `Auth/*`, `docs/M1_ACCOUNT_SECURITY.md`.

11. Are messages protected? Only the conversation's participants can read them (the Owner cannot read patient conversations). Email alerts contain no message text. Real-time channels are authorized per conversation. `Messaging/*`, `docs/M17_PRODUCTION_HARDENING.md`.

12. Is there an audit trail? Yes, append-only: sign-ins, record changes (field names only, not values), money events and assistant-answer edits. The Owner can search and export it. `Audit/*`.

13. What about RA 10173 compliance? We followed its principles (least data, access control, audit, no message text in emails, location never stored). Formal compliance needs the clinic's privacy notice, DPO review and retention decisions: listed in `docs/defense/OPEN_ITEMS.md` (item 8).

14. Do you store card numbers? No. Card and e-wallet details are entered on PayMongo's hosted page; the clinic server never sees them.

C. Payments

15. Which payment methods work? Whatever the clinic's PayMongo account enables (in our test account: GCash, card and others). The list comes from PayMongo at run time. Test mode only in this project.

16. What if PayMongo is down? The patient pays at the clinic as today; the invoice stays "Payment due" and Staff record the payment. A late webhook is still accepted and settles once.

17. Partial payments and refunds? Not online. Online payment is for the full balance; partial payments and refunds stay at the front desk (OPEN_ITEMS 11).

D. Technology and scalability

18. Why Laravel and React? Laravel gives server-side validation, database transactions, queues, mail and authentication out of the box; React with Vite gives a fast, phone-friendly portal. Both are widely taught and hosted cheaply. PostgreSQL enforces the money rules (receipts cannot be edited) at the database level.

19. Can it handle three branches and more? Branches are data, not code: the Owner adds or renames a branch and it appears in booking, the website, emails and the assistant. The clinic sees 10–20 patients a day (Q7); the design is not near any limit at that volume. We have not run a formal load test.

20. What if Gmail is down or hits its daily limit? Emails wait in the outbox and are retried automatically (up to 4 attempts over about 20 minutes); in-app notifications still work. Gmail limits daily sending for personal accounts; for production we would move to a transactional email provider using the clinic's own domain. `docs/M18_EMAIL_DISPATCH.md`.

21. How is it deployed? One installation per clinic (white-label: name, colours, logo and contact set by the Owner), on a small cloud server with PostgreSQL, a queue worker and the scheduler, with the clinic's own PayMongo account and email domain. `docs/WHITE_LABEL_CHECKLIST.md`.

22. How does it compare with the clinic's current software (about ₱1,400 a month, Q14)? The clinic likes its current software because it is centralized (Q18). Ours adds what the interview asked for: online booking without conflicts, online payment with automatic receipts, front-desk messaging and an assistant for routine questions (Q23). We do not claim it replaces their records software today.

23. What would it cost the clinic? See `docs/defense/PITCH.md`: a pricing **proposal**, not a quote. Running costs are hosting, email and PayMongo's per-transaction fees, which the clinic pays to PayMongo directly.

24. Who supports it? Q31: the clinic has reliable internet; the person for day-to-day troubleshooting was not named. We propose a support contract with a named contact and the pre-defense style health check (`scripts/predefense-check.sh`).

E. Hard questions

25. Which modules are not finished? Not started: M3 Work Schedules, M10 Dental Charts, M12 Treatment Plans, M15 Leave Management, M24 Lab Cases, M25 Patient Handovers. Mock only (browser-local): M19 Prescriptions, M20 Patient Follow-Up. Partly built: M1, M2, M4, M8, M9, M13, M14, M16, M21. Full table: `AUTOMATION_EVIDENCE.md`.

26. Why did you remove modules from the patient portal? HMO, Prescriptions, Referral & Loyalty, Follow-Up Care and Live Queue were removed from the patient portal because they had no real server data behind them; we show only what is real. HMO is out of the new scope (`MODULE_COVERAGE.md`).

27. Is there a known bug? Yes: some timestamps written by the framework are stored 8 hours early because the app and the database disagree on the time zone. Screens that use the clinic clock are correct; we fixed the new tables. The plan to fix the rest safely is in `docs/TIMEZONE_PLAN.md` (after the defense, with a backup and rollback).

28. Are the invoices BIR-compliant? Not confirmed. Optional BIR fields per branch exist and print on the PDFs, but the clinic's accountant must confirm the requirements (OPEN_ITEMS 10).

29. There's only one Owner account. Isn't that a risk? Yes; additional administrator accounts are an open decision (OPEN_ITEMS 12). Employee accounts are created by invitation.

30. Are the branch details real? Yes: names, addresses, phones, emails and hours were given by the clinic (Monday to Friday 8:00 AM to 8:00 PM, Saturday 8:00 AM to 5:00 PM, Sunday closed). Two things are still open: branches 1 and 2 have the same phone number (to be confirmed), and branch 3 has no email and an incomplete address (OPEN_ITEMS 1–2). Online times also depend on the dentists' schedules, which are still project values (OPEN_ITEMS 3).

31. Are the prices real? The demo prices are the lower end of the ranges in Q12 and are labelled as demo prices. Patients never see prices online; the clinic confirms them.

32. Does the clinic actually send reminders through the system? Not yet. PI-31 says they remind patients for every appointment; the system sends confirmation and change emails but no scheduled reminder (OPEN_ITEMS 14).

33. Are the hours hard-coded anywhere? No. Each branch stores its hours per weekday; booking, the calendar, the website, emails and the assistant all read them, so the Owner's change in Administration → Branches applies everywhere at once (`backend/tests/Feature/Appointments/BranchHoursTest.php`).

34. What did you test? The full backend suite (PHPUnit, PostgreSQL), the frontend unit tests, a render smoke test of every screen, a production build, and browser walk-throughs on phone and desktop with screenshots (`docs/ui-review/`).

F. The process change (see PROCESS_DOCUMENTATION.pdf)

35. Why this flow? It follows what the clinic told us: the receptionist books, checks in, bills and follows up (Q27), conflicts are the most frequent error (Q20) and the Owner answers Messenger herself (PI-43). So the system moves booking to the patient with only real open times, routes messages to the branch front desk, and makes billing follow the treatment. Each AS-IS step in the document cites the interview; steps we could not confirm are marked "to confirm" (SECOND_INTERVIEW_QUESTIONS.md).

36. What was removed from the old process? Typing bookings and checking for conflicts by hand (the server and the database refuse overlaps), typing online payments (PayMongo's verified webhook settles them), preparing receipts (emailed PDF), and the Owner answering routine questions (the Clinic Assistant answers from the clinic's own answers). Reminders are not automated yet (PI-31).

37. What if the internet is down? The portal needs the internet. The front desk keeps working the same way as before (phone, counter payment) and records it later in the system; emails wait in the outbox and are retried automatically (up to 4 attempts) when the connection returns. Q31: the clinic says its internet is reliable.

38. Who can see what? Patients see only their own records; Staff see their branches; the Dentist sees the patients they treat; the Owner oversees the clinic but cannot read patient conversations. Every change is in the append-only audit trail (MODULES_FINAL.md, M1, M17, M22).

39. How is the data protected? Server-side role checks on every request, email codes for sign-in, passwords stored as hashes, card details only on PayMongo, append-only financial and audit records, soft delete or retraction instead of hard delete, and knowledge mode for the assistant (nothing typed leaves the system). Retention periods and the DPO review are still open (OPEN_ITEMS 8, 16).

G. Automations (Administration → Automations)

40. What if a reminder email fails? Every email is written to the outbox first and sent by the worker; a failure is retried automatically (up to 4 attempts, with a growing wait) and recorded. The Owner sees sent, skipped and failed counts for the last 7 days on each automation card, and the pre-defense check fails if an email is stuck. A patient without a verified email is skipped, not guessed.

41. What if it sends twice? It cannot by design: each action has one key per appointment, invoice or visit and type (for example "reminder for appointment X at revision 2"). The key is unique in the database for both the automation log and the email outbox, so a retry, a second scheduler or a crash-restart finds the key already used. Just before sending, the outbox re-checks the record: a cancelled, moved or paid record is not emailed. Tested in `Automations/AutomationsTest`.

42. Who can turn it off? Only the Owner, per automation, in Administration → Automations; every switch and setting change is in the audit trail with who, the old and the new value. Since interview round 2 every automation **ships off until the Owner approves the wording** of its emails (R2-09), and recall also needs a Dentist (R2-11). Each default says whether it came from the clinic (R2 answer number) or is our recommendation.

43. Is any message sent without human approval? No wording is sent that the Owner did not approve; editing the wording clears the approval and switches the automation off. The emails only repeat facts already in the records (a time, a balance, a count), never clinical content. One automatic decision exists, because the clinic asked for it: after 15 minutes without check-in the time is released (R2-10). It is a setting ("release automatically" or "ask the front desk"), and the front desk can undo a release the same day.

44. What is your late-arrival policy, and who decided it? The clinic gave the numbers and asked us to suggest the flow (R2-07, R2-10). At 5 minutes the patient gets "Are you on your way?" (in the app and by email, with "On my way" and "Reschedule"; the reply shows on the Check-In page). At 10 minutes they show as Late; if they then check in, they go after the on-time patients in that dentist's queue (R2-07), and Staff can restore their place with a reason. At 15 minutes the time is released, recorded as a no-show, and a rebook email goes out (R2-10). Every minute is an Owner setting; 5 is our recommendation, 10 and 15 are the clinic's. Tested in `Automations/InterviewRoundTwoTest`.

45. What happens to a freed slot? It is offered, never held: first to patients booked later the same day at that branch whose service fits ("An earlier time is available"); the first to accept gets it through the normal scheduling checks, and the database's no-double-booking constraint is the final guard, so two patients accepting at once cannot both get it (the second sees "another patient took this time"). After a head start the waitlist is told, and the front desk sees the free time for walk-ins. Nobody is moved without accepting, and offers expire when the time passes. The Owner summary counts how each freed time was refilled.

46. Why can't patients book braces directly? The clinic said so (R2-17): braces and other sensitive services start with a consultation, and that consultation is what the patient books and pays for. Each service has an Owner setting: bookable, "starts with a consultation" (pointing to the consultation service) or not shown online. The server refuses a direct online booking, not only the screen; the front desk can still book it.

47. Do you process HMO claims? No. The clinic accepts some HMOs (R2-16), but the list it sent contained web-citation markers, so it is loaded **not verified** and hidden from patients until the Owner confirms each name. Patients may add their HMO provider and member number (stored encrypted) for the front desk to check at the visit. There is no coverage calculation and no claim submission in this release.

48. Where does the ID photo go? Nowhere. "Scan your ID" is optional and needs consent first. The photo is read in the patient's own browser (the PhilID QR code, a passport's machine-readable lines, or on-device OCR whose files the clinic site serves itself) and cleared right after. Only the name and birth date fill the form, highlighted for the patient to check. No image and no ID number is uploaded or stored; the server even refuses those fields. A test searches every table, the log and the email outbox after a scan-and-register run and finds nothing. Staff check the physical ID once and we store only the ID type, who checked it and when.

49. What about minors? A patient under 18 (computed in Philippine time) cannot self-register: a parent or guardian adds them as a dependent, in the counter form or the portal, with consent recorded per dependent. The child needs no ID or email; the guardian's ID is the one checked. The guardian books, is reminded and pays for the child ("Booking for: Me / Ana"). Staff see "Minor: guardian [name]". At 18 the guardian or the clinic invites the child to claim their own account; once they verify it, the guardian's access ends (audited) and the history stays with the patient.

50. What stops someone joining the queue from home? Joining today's queue needs the code shown on the front-desk screen. It changes every clinic day and staff can refresh it at any time (the old one stops working). The public endpoint is rate-limited per IP and per device, and every registration still waits for staff to check the ID before the patient is admitted. The poster's QR only opens the form; it carries no code.

51. How do you avoid duplicate patients across branches? R2-08: each branch had its own database, so patients re-registered when they moved. Now there is one shared record. Before a counter registration becomes a patient, staff see possible matches from any branch (same name and birth date plus the same phone or email) as "Record found from Branch X" and choose "Use this record" or create a new one. Nothing is merged automatically, and the public page never reveals whether a record exists.

52. What stops someone taking over my record? Knowing your name and birth date is not enough. The claim code goes only to the email already on your record, never to an address the claimant types, and the answer looks the same whether or not anything matched. If your record has no email, only the front desk can offer you an online account, after checking your ID in person and confirming your email with you; the link works once and expires. Your password is chosen together with the emailed code, so nobody who can't read your inbox can set it. Codes die after 5 wrong tries and every claim is audited. `Intake/IntakeTest` (claim tests).

53. How do you handle a dentist's leave? M15. The dentist asks in the system (vacation, sick or other, full or part days; a private note only the approver sees). The Owner approves, or also the branch front desk if the Owner allows it; nobody approves their own leave. Before approving, the Owner sees the booked visits in that period with a suggested cover dentist (same branch, same service, free at that time). Approving removes the dentist's time from booking at once, but cancels nothing: the front desk moves each visit to the cover dentist in one click or emails the patients a reschedule link. Cancelling the leave makes the time bookable again. We do not count leave balances: the clinic gave no policy. `Staffing/LeaveTest`.

54. What if a schedule changes with booked patients? M3. Every booking rule reads one source: the branch's opening hours, the dentist's weekly hours or that day's exception, the break and approved leave. Hours outside the branch's opening hours are refused. A change that leaves booked visits without their dentist never cancels them: the system lists them with a suggested cover dentist and one action to email the patients a link to choose a new time, in wording the Owner approved. Every change is audited with the old and new hours. `Staffing/WorkSchedulesTest`.

55. Who sees a handover? M25. The sending side (the front desk or the treating dentist) and the receiving side (the receiving branch's front desk and the receiving dentist). The receiving front desk gets the patient's record through the normal branch rule while the handover is open and for 60 days after it is acknowledged (an Owner setting); the receiving dentist sees the handover itself. The Owner sees only who, when, from and to. The patient is told neutrally that the record was shared, without clinical text.

Every create, read by the receiving side and acknowledgement is audited, and the audit holds no clinical text. `Staffing/HandoverTest`.